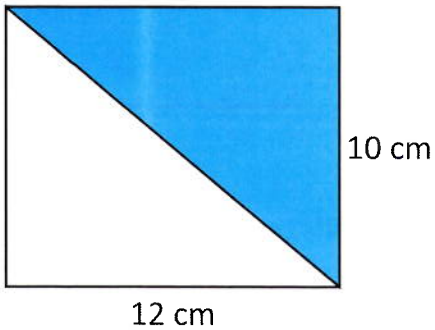
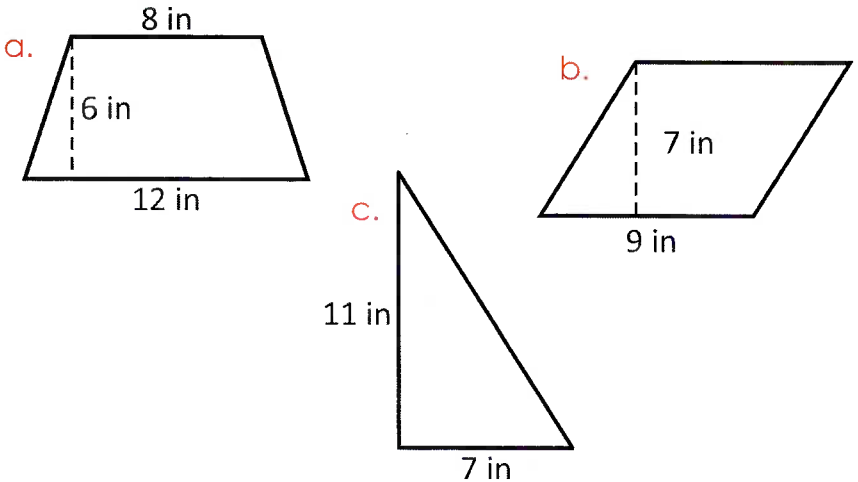
Directions: Determine the area of the blue triangle.



Directions: Determine the area of each shape, and put them in order from least to greatest.



_____ < _____ < _____

Level 5b

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Name: Ojas

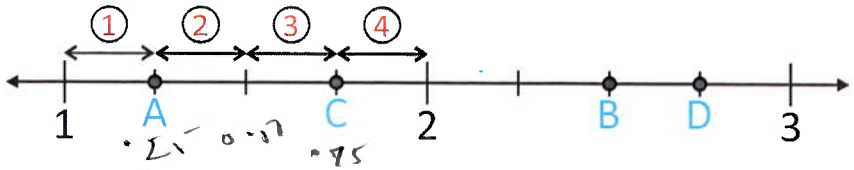
Date: / /

Time: ____ : ____ TO ____ : ____

Week 3 C1

Decimals on a Number Line | **Scientific Notation**

To determine the values in between two whole numbers on a number line, count how many spaces are between the numbers.



There are 4 spaces between 1 and 2. To find the value of each space, divide 100 by the number of spaces.

100 ÷ 4 = 25

Put a decimal before the number = .25

That is how much each space is worth.

Find the value of each letter.

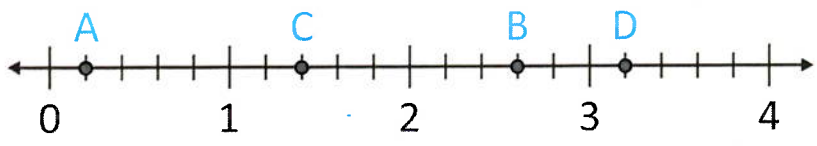
A = 1.25

C = 1.75

B = 2.50

D = 2.75

Directions: Write the correct decimal next to each letter.



There are 5 equal spaces between 0 and 1.

100 ÷ 5 = 20

Each tally mark equals 0.20

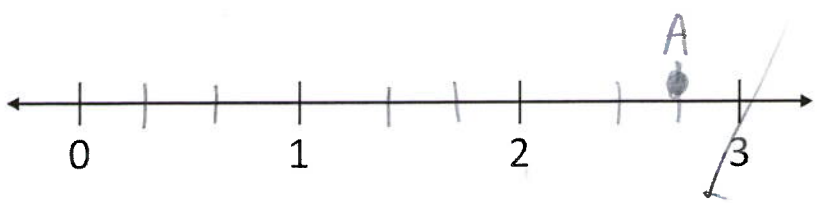
A = .20

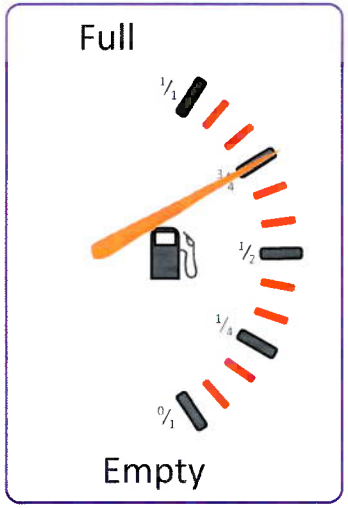
C = 1.40

B = 2.60

D = 3.20

Directions: Divide the number line into thirds. Plot $2.\overline{66}$ with an A.





What fractional amount represents the amount of fuel in the gas tank?

_____

Decimal: _____

Percent: _____

Directions: Complete the table.

Fraction	Decimal	Percent
	.5	
		$33.\overline{3}\%$
$\frac{3}{5}$		

Directions: Complete the chart.

352.02	Written Form _____ _____
	What number is in the hundredths place? _____

four thousand, six hundred twelve and twenty-eight hundredths	Standard Form
What number is in the tenths place? _____	

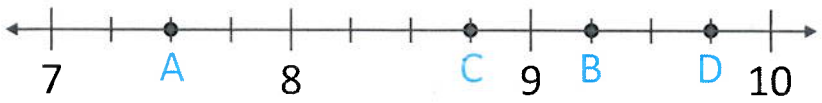
Directions: Insert <, > or =.

1) twenty-seven and four hundred fifteen thousandths ☐ twenty-seven and five hundred thousandths

2) 1,458.093 ☐ 1,399.99

GM5bC7.6

Directions: Determine the value of each letter.



A = 7.50 C = 8.75

B = 9.25 D = 9.75

Directions: Solve.

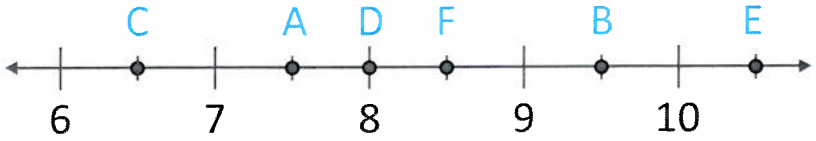
Zooey swam 2 laps in forty-three and thirty-two hundredths of a second.

Write her time as a decimal and estimate roughly where it belongs on the number line below.

43.32
Decimal

GM5bC1.3

Directions: Write the correct decimal next to each letter.



There are 2 equal spaces between 6 and 7.

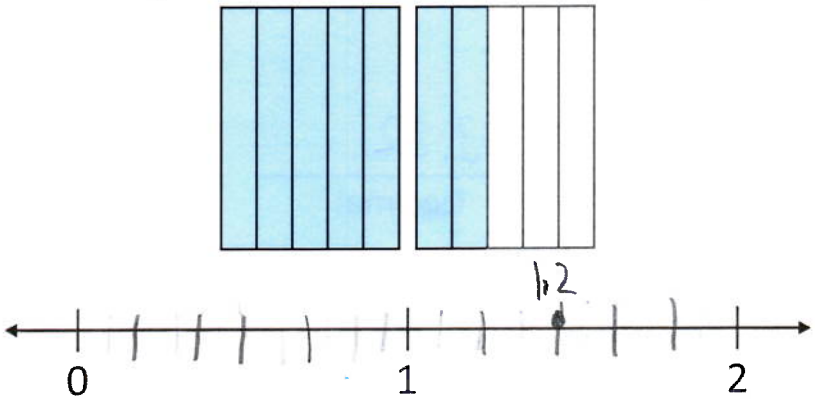
100 ÷ 2 = 50

Each tally mark equals 0.50

A = 7.50 C = 6.50 E = 10.50

B = 9.50 D = 8.00 F = 8.50

Directions: Represent the decimal on the number line.

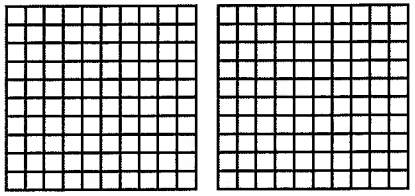


Directions: Explain in words how many 100 grids you will need to represent each number.

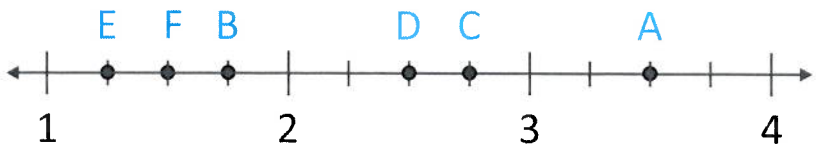
2.63

8.49

Directions: Complete the chart.

Decimal	Grid
1.5	
Written Form	

Directions: Write the correct decimal next to each letter.



There are [ ] equal spaces between 1 and 2.

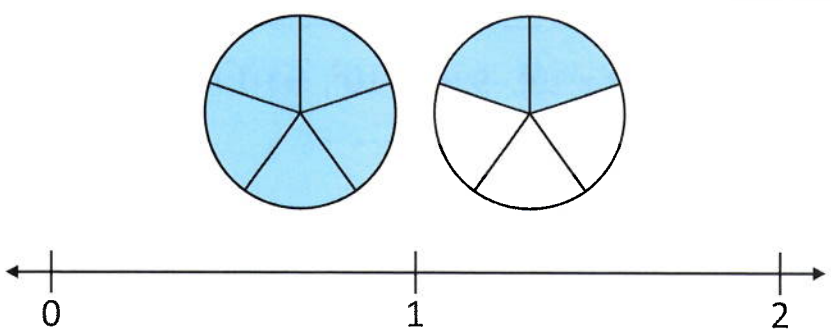
Each tally mark equals [0.]

A = _____ D = _____

B = _____ E = _____

C = _____ F = _____

Directions: Represent the decimal on the number line.



Scientific Notation

Scientific notation is a way to write really large and really small numbers. This form of expressing numbers is commonly used in science.

For example, the sun is approximately 92,960,000 miles from Earth. This can be expressed as 9.296×10^7.

Step 1: Move the decimal point so that there is only one **non-zero** digit to the left of the decimal.

92960000 = 9.2960000

Step 2: Count the number of decimal places the decimal "moved" from the original number. This number becomes the exponent of 10.

9.2960000
7 6 5 4 3 2 1

At this point, we can remove the extra zeroes since they do not affect the number.

Step 3: Let's put it all together.

$92960000 = 9.296 \times 10^7$

number of times decimal moves

base ten

original number rewritten with one digit in front of decimal

Directions: Circle all the expressions in correct scientific notation.

26.00×10^2

1×10^3

3.279×10^8

$5{,}000 \times 10^2$

0.034×10^5

7.29×10^2

Directions: Express the number in scientific notation.

On average, a humming bird beats its wings 1,036,800 times in 3 days.

1.0368×10^6

$1{,}0368 \times 10^6$

Directions: Match each number in scientific notation to its correct standard form.

8.21×10^3	82.1
8.21×10^1	821,000
8.21×10^5	82,100,000
8.21×10^7	8,210

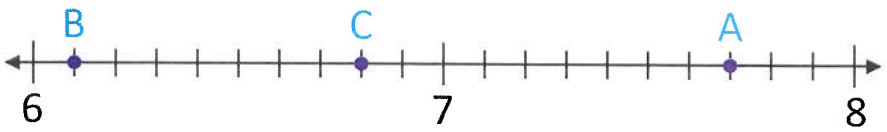
Directions: Order from least to greatest.

4.54×10^6, 4.54×10^1, 4.54×10^2

_____ < _____ < _____

9.01×10^1, 9.01×10^7, 9.01×10^2

_____ < _____ < _____

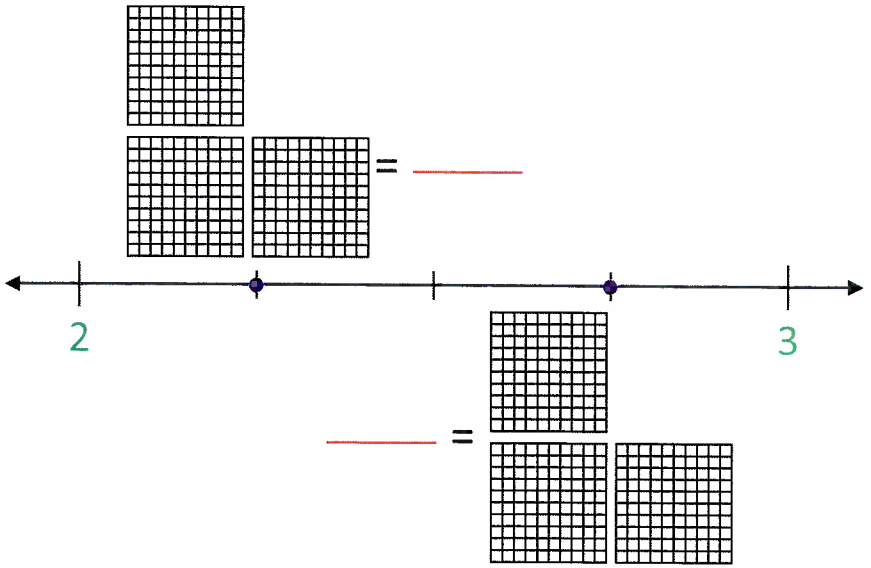


1. What is the value of A? _____

2. What is the value of B? _____

3. What is the value of C? _____

Directions: Shade each box to represent each number as a decimal.



GM5bC7.2

Directions: Express in standard form.

1. $3.2782 \times 10^4 =$ 3,278,200
2. $5.210 \times 10^2 =$ 521,000
3. $8.9 \times 10^7 =$ 890,000,000

Directions: Express in scientific notation.

1. 279,381 = $279{,}381 \times 10^5$
2. 48.34 = $4{,}834 \times 10^2$ 4.834×10^1
3. 566.3 = $5{,}663 \times 10^1$ 5.663×10^2

GM5bC1.7

Directions: The table shows each planet's approximate distance from the sun. Complete the table.

Planet	Kilometers from the sun	Scientific Notification
Mecrury	57,900,000	5.79×10^7
Venus	108,200,000	1.082×10^8
Earth	149,600,000	1.496×10^7
Mars	227,900,000	2.279×10^8
Jupiter	7,784,000,000	7.784×10^8
Saturn	1,429,400,000	1.4294×10^9
Uranus	2,875,000,000	2.875×10^9
Neptune	4,504,300,000	4.5043×10^9

Level 5b

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 3

C7

Level Review

Directions: Find the missing length. Show your work.

1. Find the **height** of a trapezoid with bases 8 inches and 6 inches and area of 28 inches^2.

2. Find the **base** of parallelogram whose height is 20 cm and area is 300 cm^2.

Directions: Express each of the following durations in standard form.

1. The Great Pyramid of Giza is approximately 4.575×10^3 years old.

   _____

2. Stonehenge is approximately 50.16×10^2 years old.

   _____

3. The Great wall of China is approximately 2.221×10^3 years old.

   _____

Directions: Circle the greatest number.

4.261×10^4 418,932

5,210,364

39.86×10^5 7.32×10^2

Level 5b

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 3

C2

Decimals on a Number Line

Scientific Notation

_____ _____.____ _____ _____

(1) Double 19 and then subtract the number of days in December. Write the answer in the thousandths place.

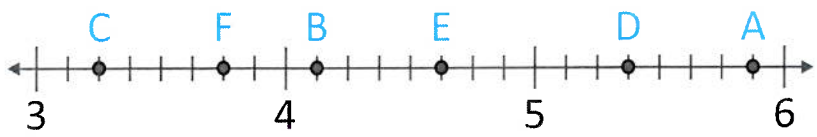
(2) Divide 63 by the number of days in a week. Write the answer in the tens place.

(3) Subtract 65 from the number of days in a year. Then divide by 60. Write the number in the tenths place.

(4) The number in the ones place is 3 less than the number in the tens place.

(5) Multiply the number of weekdays by the number of days in the weekend. Then subtract 6. Write this number in the hundredths place.

Directions: Write the correct decimal next to each letter.



A = _____ D = _____

B = _____ E = _____

C = _____ F = _____

Directions: Divide the number line and graph the points below.

What increment should you use for each tally mark?

_____

A = 1.3 C = 2.5 E = 2.8

B = 1.9 D = 1.6 F = 2.2

GM5bC2.2

Directions: Determine if all the numbers represent the same value. Explain your answer.

6.243×10^3, 624.3×10^3, 6,243

Are these equal? Yes / No

Explain _____

_____

_____

_____

Directions: Express the following numbers in scientific notation.

1. The circumference of Pluto is approximately 7,232,392 meters.

_____

2. It takes Pluto approximately 90,520 days to orbit the sun.

_____

GM5bC6.7

Directions: Match each number in scientific notation to its correct standard form.

5.26×10^4	526
5.26×10^2	5,260,000
5.26×10^6	526,000,000
5.26×10^8	52,600

Directions: Order from least to greatest.

4.21×10^4, 4.21×10^6, 4.21×10^3

_____ < _____ < _____

8.79×10^8, 8.79×10^4, 8.79×10^1

_____ < _____ < _____

GM5bC6.6

Directions: Place each letter on the number line.

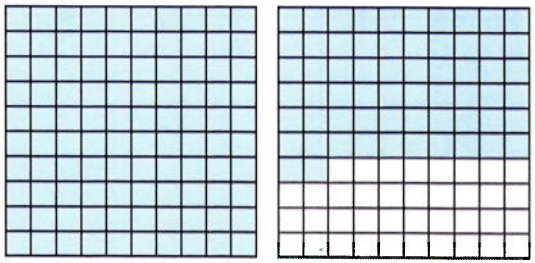
1

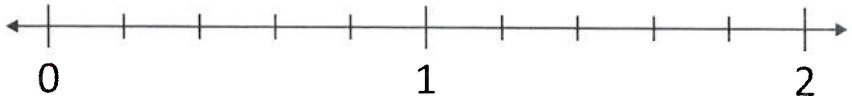
A = $2.\overline{33}$ B = $3.\overline{66}$ C = $1.\overline{33}$

2

A = 6.20 B = 8.6 C = 7.80

Directions: Represent the decimal on the number line. Then write the decimal in word form on the line.

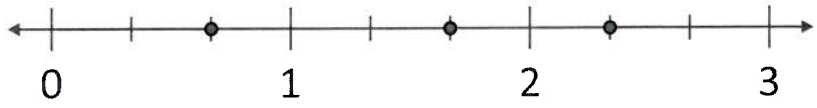




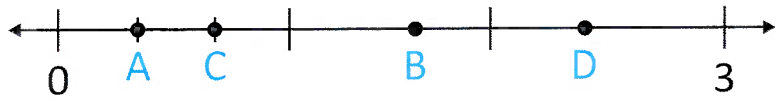
GM5bC2.3

Directions: Circle the point that has been placed incorrectly on the number line. Then explain the error.

$.\overline{33}$, $1.\overline{66}$, $2.\overline{33}$



Directions: Choose the values that can be true for each letter. Circle all that apply.



1) A = .75 C = 1.1
B = 2.15 D = 2.8

2) A = .5 C = .6
B = 2.2 D = 3.1

3) A = .5 C = .6
B = 1.75 D = 2.5

GM5bC2.4

Scientific Notation

Directions: Simplify and insert <, > or =.

1) 3.62×10^3 ☐ 41.3×10^2

2) 81.23×10^2 ☐ 80.95×10^2

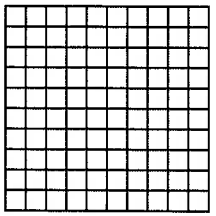
Directions: Use the chart to answer the questions.

Sample	Measurement (meters)
CN Tower (height)	2.178×10^3
Grand Canyon (depth)	731.16×10^2
Mississippi River (width)	696.96×10^3

Which of the above samples recorded the smallest measurement?

GM5bC6.5

Directions: Complete the chart.

Decimal	Grid
.65	

Number Line

Directions: Identify and circle the number line where **A** represents .85.

1

2

3

Scientific Notation

Directions: Write each number in scientific notation.

(1) 4,800 = _____ x $10^{_}$

(2) 860.45 = _____ x $10^{_}$

(3) 730,000 = _____ x $10^{_}$

(4) 10,230 = _____ x $10^{_}$

(5) 28,190 = _____ x $10^{_}$

Directions: Determine if the numbers have been written correctly in standard form.

1. $8.726 \times 10^4 = 87{,}260$ ☐ Yes ☐ No
2. $6.23 \times 10^6 = 623{,}000{,}000$ ☐ Yes ☐ No
3. $1.068 \times 10^9 = 1{,}068{,}000{,}000$ ☐ Yes ☐ No

Directions: Express in standard form.

1. $9.25 \times 10^1 =$ _____

2. $6.433 \times 10^7 =$ _____

Directions: Express in scientific notation.

1. 435,500,000 = _____

2. 13,881 = _____

Directions: Circle all the numbers that represent the same value.

3.29×10^2 $.329 \times 10^2$

329

3.29×10^4 32.9×10

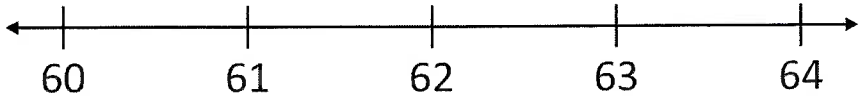
1. How many fifths are in 1? Explain.

_____

_____

_____

2. In Miami, the average annual rainfall is 61.9 inches. The average in West Palm Beach is 63.2 inches. Graph and label both cities on the number line.



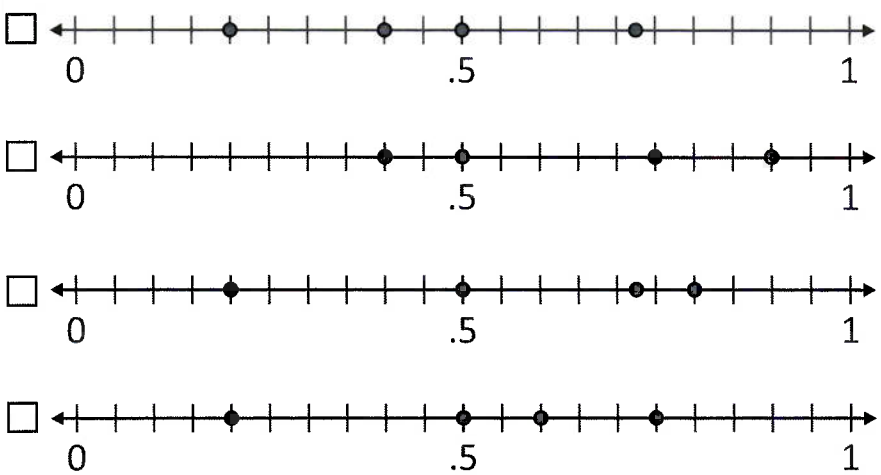
3. Graph each amount on the number line. Xandu is 33.4 inches tall, Brody is 35.3 inches and Octavia is 36.25 inches tall.

Directions: What is the distance between point A and point B on the number line?

_____

Directions: Mark the graph that correctly shows the following points.

.5, .75, .4, .9



GM5bC6.2

Directions: Use the table to determine which department had more sales.

Department	Sales ($)
Toys	3.246×10^5
Clothes	4.18×10^4
Shoes	3.89×10^5

a. Which department sold the most?

_____

b. Which department sold the least?

_____

Directions: Express each of the following measurements in scientific notation.

1. average weight of a rhino: 2,313 kg

_____

2. the height of the CN Tower: 55,321 cm

_____

3. average depth of the Dead Sea: 653 feet

_____

GM5bC2.7

Directions: Simplify and insert <, > or =.

1. 8.29×10^4 ☐ 7.893×10^3

2. 4.63×10^2 ☐ 5.3×10^3

Directions: Use the chart to answer the questions.

Sample	Measurement (meters)
Empire State Building	3.81×10^2
Diameter of the moon	3.4748×10^6
Mount Everest	2.9029×10^4

Which of the above samples recorded the largest measurement? How did you determine that?

_____

_____

_____

_____

Level 5b

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 3

C6

Decimals on a Number Line | **Scientific Notation**

Directions: Draw a number line to solve.

At a concession stand, a hot dog costs $\frac{3}{4}$ of a dollar, a candy bar costs $\frac{3}{5}$ of a dollar, a snow cone costs \$0.70 and a piece of fruit pie costs \$0.85.

Which item costs the most?

Directions: Solve and explain.

Megan correctly wrote a problem using scientific notation.

8,310,000 ÷ 620,000

Which did Megan write? Explain your answer.

- ○ $\frac{8.31 \times 10^7}{6.2 \times 10^5}$
- ○ $\frac{8.31 \times 10^7}{6.20 \times 10^4}$
- ○ $\frac{8.31 \times 10^6}{6.2 \times 10^6}$
- ○ $\frac{8.31 \times 10^6}{6.20 \times 10^5}$

Level 5b

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Name

Date / /

Time ____ : ____ TO ____ : ____

Week 3 C3

Decimals on a Number Line | **Scientific Notation**

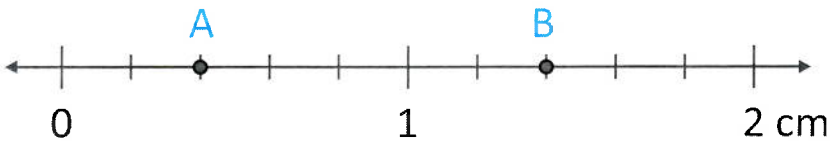
Directions: Follow the steps.

(1) Plot 93 through 100 with increments every two-tenths.

(2) The normal body temperature of a human is ninety-eight and six-tenths degrees. Plot this with a T.

(3) Plot 93.6 with a B.

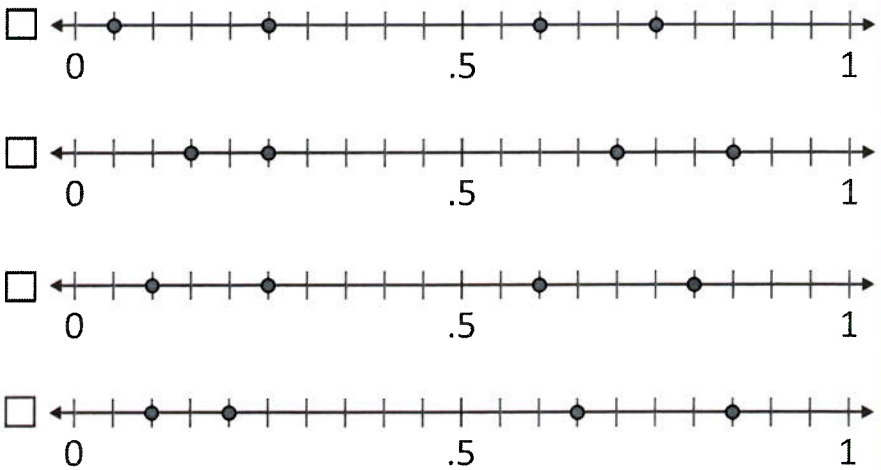
Directions: What is the distance between point A and point B on the number line?



_____

Directions: Mark the graph that correctly shows the following points.

.25, .6, .8, .1



GM5bC3.2

Directions: Express in standard form.

1. $8.95 \times 10^4 =$ _____

2. $4.63 \times 10^6 =$ _____

Directions: Express in scientific notation.

1. 82,131,500,000 = _____

2. 26,905 = _____

Directions: Circle all the numbers that represent the same value.

57.13×10^2 $.5713 \times 10^4$

5,713

5.713×10^4 5.713×10^5

GM5bC5.7

Directions: Use the table to determine which club raised the most money.

Club	Money ($)
Cultural Club	2.3450×10^4
Mathletes	3.01×10^4
Spirit Club	42.5×10^2

a. Which club sold the most?

_____

b. Which club sold the least?

_____

Directions: Express each of the following measurements in scientific notation.

1. distance from the east coast to the west coast: 2,092 miles

_____

2. the lowest temperature ever recorded: -128.6°F

_____

3. the height of Mt. Kilimanjaro: 19,341 feet

_____

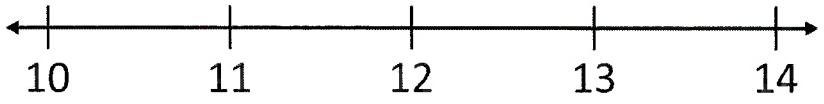
1. How many tenths are in 1? Explain.

_____

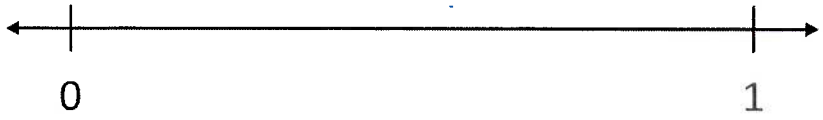
_____

_____

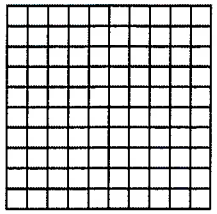
2. In Illinois, the average wind speed is 10.4 miles per hour. In Montana, the average wind speed is 13.1 miles per hour. Graph and label both states on the number line.



3. Matrix has $0.40, John has $\frac{4}{5}$ of a dollar, Lynn has $\frac{3}{10}$ of a dollar and Valerie has $0.55. Who has the greatest amount of money? Graph each amount on the number line.



Directions: Complete the chart.

Decimal	Grid
.8	

Number Line

Directions: Identify and circle the number line where **A** represents .2.

1

2

3

Scientific Notation

Directions: Write each number in scientific notation.

(1) 270 = _____ x $10^{_}$

(2) 450,000 = _____ x $10^{_}$

(3) 6,900 = _____ x $10^{_}$

(4) 9,110 = _____ x $10^{_}$

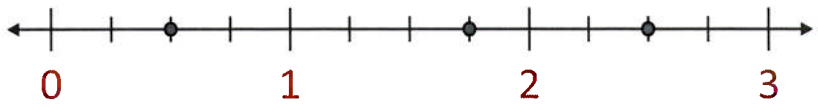
(5) 81,000 = _____ x $10^{_}$

Directions: Determine if the numbers have been written correctly in standard form.

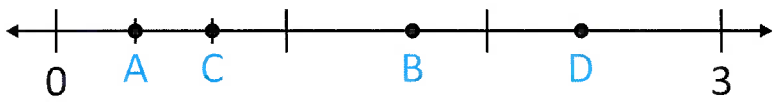
1. $36.24 \times 10^3 = 36{,}240$ ☐ Yes ☐ No
2. $513.29 \times 10^2 = 5.1329$ ☐ Yes ☐ No
3. $8.95 \times 10^6 = 895{,}000$ ☐ Yes ☐ No

Directions: Circle the point that has been placed incorrectly on the number line. Then explain the error.

1.75, .50, 2.25



Directions: Choose the values that can be true for each letter. Circle all that apply.



1) A = .75 C = 1.1
B = 2.15 D = 2.8

2) A = .5 C = .6
B = 2.2 D = 3.1

3) A = .5 C = .6
B = 1.75 D = 2.5

GM5bC5.4

Scientific Notation

Directions: Complete the chart.

Number	Standard Form	Scientific Notification
one trillion	1,000,000,000,000	
one billion		1×10^9
one million	1,000,000	
one hundred thousand		1×10^5
ten thousand	10,000	
one thousand	1,000	
one hundred	100	
ten		1.10^1

GM5bC3.5

Directions: Match each number in scientific notation to its correct standard form.

7.65×10^3	76.5
7.65×10^1	765,000
7.65×10^5	76,500,000
7.65×10^7	7,650

Directions: Order from least to greatest.

3.69×10^3, 3.69×10^1, 3.69×10^2

_____ < _____ < _____

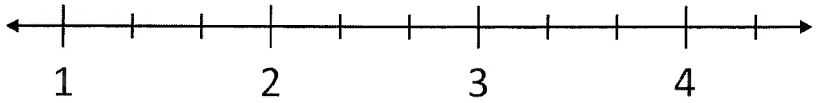
5.1×10^4, 5.1×10^5, 5.1×10^2

_____ < _____ < _____

Directions: Place each letter on the number line.

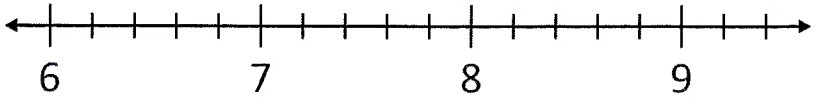
1

A = $1.\overline{33}$ B = $2.\overline{33}$ C = $4.\overline{33}$

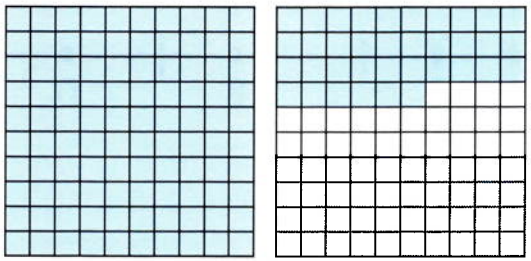


2

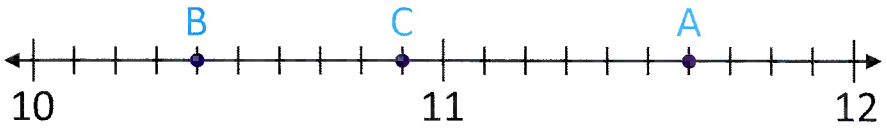
A = 6.80 B = 9.20 C = 7.40



Directions: Represent the decimal on the number line. Then write the decimal in word form on the line.



_____

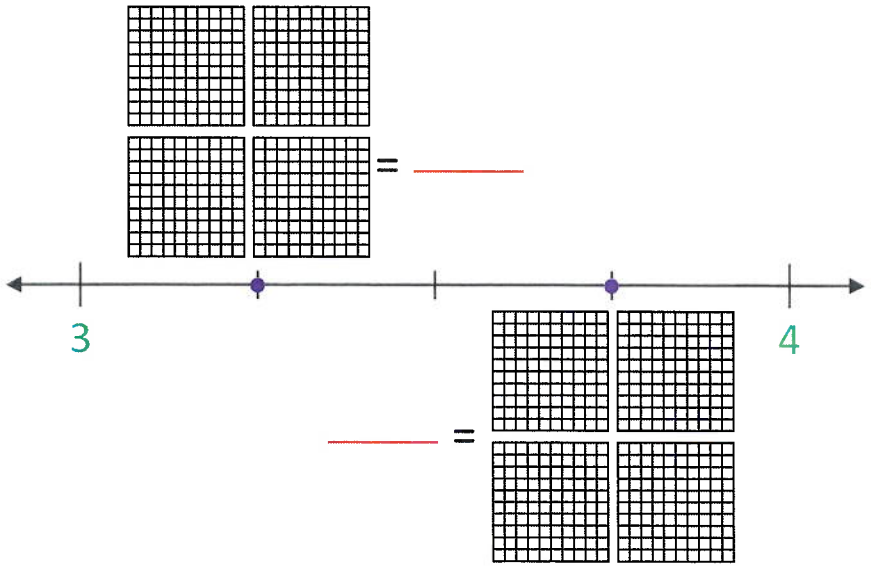


1. What is the value of A? _____

2. What is the value of B? _____

3. What is the value of C? _____

Directions: Shade each box to represent each number as a decimal.



Directions: Determine if all the numbers represent the same value. Explain your answer.

8.901×10^4, 89.01×10^3, 89,010

Are these equal? Yes / No

Explain _____

_____

_____

_____

Directions: Express the following numbers in scientific notation.

1. Venus' distance from the sun is 6,724,000 miles.

_____

2. Jupiter is about 778,120,000 miles from the sun.

_____

Directions: Express each of the following durations in standard form.

1. The Earth is approximately 4.5×10^9 years old.

_____

2. Life on Earth began about 3.5×10^9 years ago.

_____

3. Dinosaurs roamed the earth approximately 6.5×10^7 years ago.

_____

Directions: Circle the greatest number.

3.297×10^5 1,275,000

4,093,000

5.3×10^6 2.64×10^6

Level 5b

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Name

Date / /

Time ____ : ____ TO ____ : ____

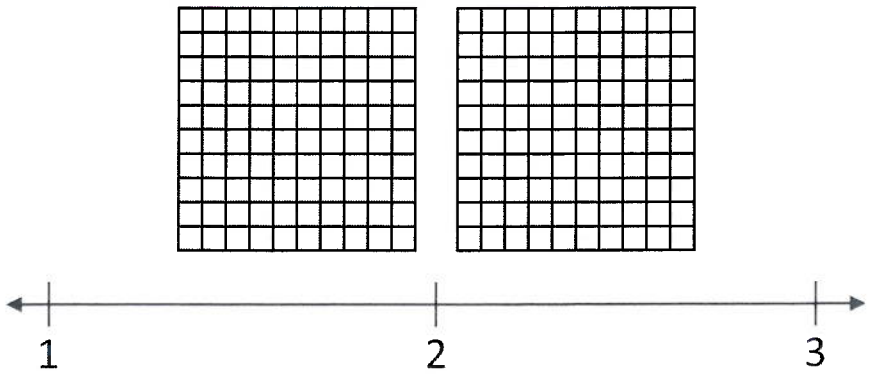
Week 3

C5

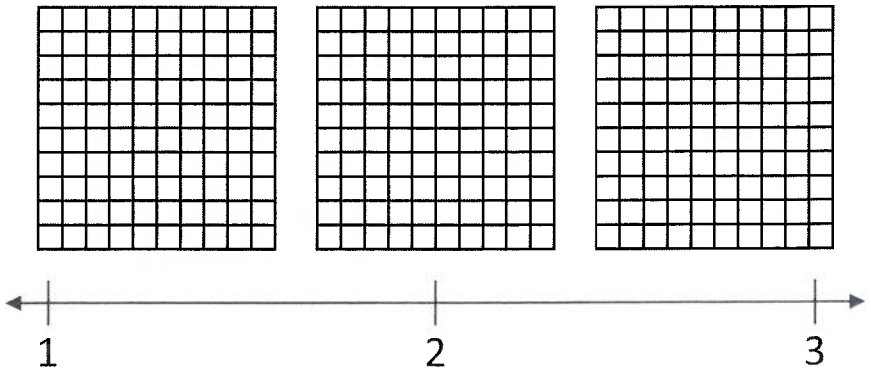
Decimals on a Number Line **Scientific Notation**

Directions: Represent each number on the grid and number line.

Derrick jogs 1.7 miles every day.



Derrick's sister walks 2.4 miles every day.



Directions: Express the following numbers in scientific notation.

1. The speed of light is 299,792,458 miles per second.

_____

2. The radius of earth is 6,378 km.

_____

3. The number of seconds in a day is 86,400.

_____

Level 5b

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 3

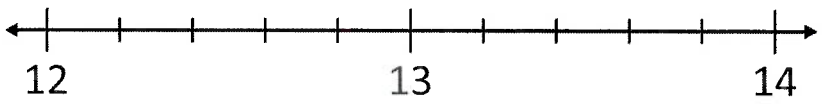
C4

Decimals on a Number Line

Scientific Notation

Directions: Answer the questions.

1. What increment is this number line divided into?

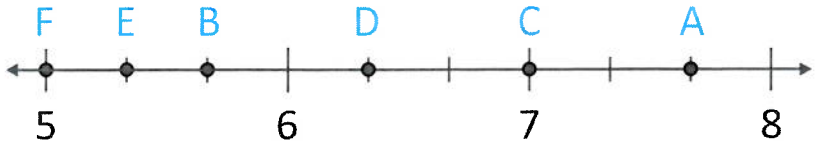


a. halves
b. thirds
c. fourths
d. fifths

2. Does point C represent 7.25?

a. Yes
b. No

Directions: Write the correct decimal next to each letter.



There are ☐ equal spaces between 5 and 6.

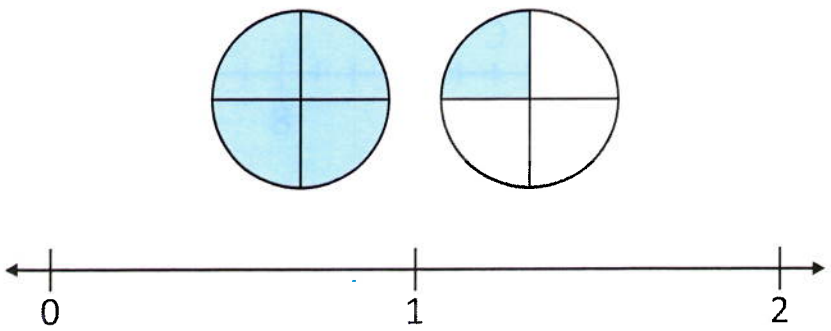
Each tally mark equals 0.

A = _____ D = _____

B = _____ E = _____

C = _____ F = _____

Directions: Represent the decimal on the number line.



Directions: Express in standard form.

1. $7.89 \times 10^4 =$ _____
2. $34.6 \times 10^9 =$ _____
3. $50.10 \times 10^2 =$ _____

Directions: Express in scientific notation.

1. 5,840 = _____
2. 695,000 = _____
3. 24.231 = _____

Directions: Circle all the expressions in correct scientific notation.

531×10^3	1×10^9
2.62×10^4	$4{,}000 \times 10^5$
0.132×10^8	8.18×10^7

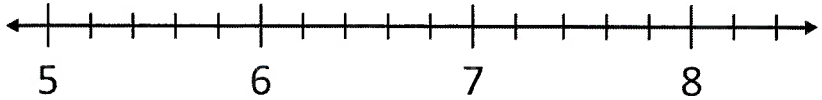
Directions: Express the number in scientific notation.

On average, a human has 100,000 strands of hair.

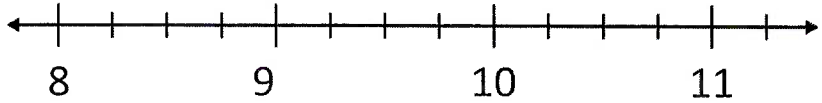
Directions: Divide the number line into fifths. Plot 1.8 with the letter B.

Directions: Place each letter on the number line.

1. A = 6.2 B = 7.6 C = 5.4



2. A = 10.50 B = 8.5 C = 9.50



Directions: Write each number as a decimal and mark it on the number line.

Lynn lives $6\frac{3}{4}$ miles from the library. Write the distance as a decimal and graph it on the number line.

Decimal = _____

Lynn's friend Mandy lives $7\frac{1}{2}$ miles from the library. Write the distance as a decimal and graph it on the number line.

Decimal = _____

Scientific Notation

1. The United States currently consumes about 19,600,000 barrels of oil a day. How is this written in scientific notation?

- ○ 1.96×10^8
- ○ 1.96×10^7
- ○ 1.96×10^9
- ○ 1.96×10^6

2. A light year is the distance light can travel in a vacuum in one year. This distance is about 9.46×10^{12} km. What is the standard form of this number?

- ○ 9,460,000,000
- ○ 94,600,000,000
- ○ 946,000,000,000
- ○ 9,460,000,000,000